



## Model 2015

(1)  $(U, V)$  - vector aleator discret

a)  $a, b = ?$ ,  $E(V) = 0,15$

| $U \backslash V$ | -1   | 1    | b   |
|------------------|------|------|-----|
| 1                | 0,25 | 0,05 | a   |
| 3                | 0,3  | a    | 0,1 |

$$E(X) = \sum_{i=1}^n x_i \cdot p_i$$

| $V$ | -1   | 1        | b       |
|-----|------|----------|---------|
|     | 0,55 | $0,05+a$ | $0,1+a$ |

| $U$ | 1                  | 3       |
|-----|--------------------|---------|
|     | $0,30+a$<br>$0,45$ | $0,4+a$ |

$$\sum_{i=1}^n P(U=i) = 1 \Leftrightarrow 2a + 0,40 = 1 \Rightarrow 2a = 0,60 \Rightarrow a = 0,30$$

$$P(V=b) = 0,25$$

$$P(V=1) = 0,20$$

$$P(V=-1) = 0,55$$

$$E(V) = 0,15$$

$$\Rightarrow 0,25 \cdot b + 0,20 \cdot 1 + (-1) \cdot 0,55 = 0,15$$

$$0,25 \cdot b - 0,35 = 0,15 \Rightarrow 0,25b = 0,50 \mid \cdot 100$$

$$\Rightarrow b = 2$$

b)  $U$  și  $V$  - independen.?

$$P(U=u, V=v) = P(U=u) \cdot P(V=v) \rightarrow \text{independen.}$$

Căutăm un contraexemplu

$$P(U=1, V=-1) = P(U=1) \cdot P(V=-1) \Leftrightarrow 0,25 = 0,45 \cdot 0,55$$

$$0,25 \neq 0,25$$

$$P(U=1, V=1) = P(U=1) \cdot P(V=1) \Leftrightarrow 0,05 = 0,45 \cdot 0,2$$

$$0,05 \neq 0,09 \text{ (F)} \Rightarrow U \text{ și } V \text{ nu sunt independen.}$$

c)  $E[(U-3)^2]$

?. ?

$$(2) \quad X \sim \begin{pmatrix} 0 & 1 & 2 & 3 \\ \frac{5^3}{6^3} & \frac{5^2 \cdot 3}{6^3} & \frac{5 \cdot 3^2}{6^3} & \frac{3^3}{6^3} \end{pmatrix} \quad 6^3 \text{ poob.}$$

$$a) \quad E(X) = \sum_{i=1}^n x_i \cdot P(X=x_i)$$

$$E(X) = \frac{5^3}{6^3} \cdot 0 + \frac{5^2 \cdot 3}{6^3} \cdot 1 + 2 \cdot \frac{5 \cdot 3^2}{6^3} + 3 \cdot \frac{3^3}{6^3}$$

$$= \frac{3 \cdot 5^2 + 30 + 3}{6^3} = \frac{108}{216} = \frac{1}{2}$$

$$b) \quad X \sim \text{Brmo}(432, \frac{1}{6^3}) \Rightarrow E(X) = 432 \cdot \frac{1}{6^3} = 2.$$

$$(3) \quad f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = \begin{cases} \frac{3a^3}{x^4}, & x > a \\ 0, & x \leq a \end{cases}, a > 0.$$

$$P(X \leq 2a)$$

$$E(X)$$

$$P(X \leq 2a) = \int_a^{2a} \frac{3a^3}{x^4} dx = 3a^3 \int_a^{2a} x^{-4} dx = 3a^3 \cdot \left( -\frac{1}{x^3} \right)$$

$$= -\frac{a^3}{x^3} \Big|_a^{2a}$$

$$= -\frac{a^3}{(2a)^3} + \frac{a^3}{a^3}$$

$$= -\frac{1}{8} + 1 = \frac{7}{8}$$

$$E(X) = \int_a^{\infty} x \cdot \frac{3a^3}{x^4} dx = 3a^3 \int_a^{\infty} \frac{1}{x^3} dx$$

$$= 3a^3 \left( -\frac{1}{2x^2} \right) \Big|_a^{\infty} = 3a^3 \cdot \frac{1}{2a^2}$$

$$= \frac{3a}{2}$$

(4)  $X, Y$  val binaire  $(0, 1)$  a)  $P(X=Y)$

$$P(X=1) = 0,4$$

$$P(X=1) = 0,4 \Rightarrow P(X=0) = 0,6$$

$$P(Y=1 | X=1) = 0,2$$

$$P(X=Y=1) + P(X=Y=0) =$$

$$P(Y=0 | X=0) = 0,3$$

$$= P(X=1) P(Y=1 | X=1) + P(Y=0 | X=0) \cdot P(X=0)$$

$$X \sim \begin{pmatrix} 0 & 1 \\ 0,6 & 0,4 \end{pmatrix}$$

$$= 0,4 \cdot 0,2 + 0,3 \cdot 0,6$$

$$= 0,08 + 0,18 = 0,26$$

b)  $E(Y) = P(Y=0) \cdot 0 + P(Y=1) \cdot 1 = P(Y=1)$

$$P(Y=1) = P(Y=1 | X=1) \cdot P(X=1) + P(Y=1 | X=0) \cdot P(X=0)$$

$$= 0,2 \cdot 0,4 + (1 - P(Y=0 | X=0)) \cdot P(X=0)$$

$$= 0,08 + 0,4 \cdot 0,6 = 0,32$$

c)  $E(X \cdot Y) = P(X=Y=1) \cdot 1 + P(X=Y=0) \cdot 0 = P(X=Y=1)$

$$P(X=Y=1) = P(Y=1 | X=1) \cdot P(X=1) = 0,2 \cdot 0,4 = 0,08$$

d)  $E(\underbrace{2^1 Y + 2^0 X}_{\text{bimon} \rightarrow B_{10}}) = 2E(Y) + E(X) = 2 \cdot 0,32 + 0,4 = 1,04$

(5)  $f_X: \mathbb{R} \rightarrow \mathbb{R}$

$$f_X(t) = \begin{cases} c(3-t)^2, & 0 < t < 3 \\ 0, & \text{altfel} \end{cases}$$

a)  $c=?$

b)  $F_X=?$

c)  $P(1 < X < 2)$

$P(X < 2 | X > 1)$

$$a) \int_0^3 c(3-t)^2 dt = 1 \Rightarrow -c \frac{(3-t)^3}{3} \Big|_0^3 \quad (2) \quad c \frac{3^{3/2}}{3} = 1$$

$$9c = 1 \Rightarrow c = \frac{1}{9}$$

$$b) F_X(x) = \int_{-\infty}^x f_X(t) dt = \begin{cases} 0, & x \leq 0 \\ \int_0^x \frac{1}{9} \cdot (3-t)^3 dt = \frac{(3-t)^3}{27} \Big|_0^x, & 0 < x < 3 \\ 1, & x \geq 3 \end{cases}$$

$$= \begin{cases} 0, & x \leq 0 \\ \frac{x^3}{27} - \frac{x^2}{3} + x, & 0 < x < 3 \\ 1, & x \geq 3 \end{cases}$$

$$\begin{aligned} c) P(1 < X < 2) &= F_X(2) - F_X(1) = \frac{2^3}{27} - \frac{2^2}{3} + 2 - \frac{1}{27} + \frac{1}{3} - 1 \\ &= \frac{8-1}{27} + \frac{1-4}{3} + 1 \\ &= \frac{4}{27} \end{aligned}$$

$$\begin{aligned} P(X < 2 \mid X > 1) &= \frac{P(1 < X < 2)}{P(X > 1)} \\ &= \frac{F_X(2) - F_X(1)}{1 - F_X(1)} \end{aligned}$$

(6) 4 bile verzi  
5 bile albastre  
6 bile roșii

$i \in \{1, 2, 3\}$

Formula înmulțirii totale

$$P(A_1 \cap \dots \cap A_m)$$

$$= P(A_1) \cdot P(A_2 | A_1) \cdot \dots$$

$$\dots \cdot P(A_m | A_1 \cap \dots \cap A_{m-1})$$

$A_i$ : „pe obiect, o bilă albăstră”

$$\begin{aligned} P(V_2 \cap R_3) &= P(V_1 \cap V_2 \cap R_3) + P(A_1 \cap V_2 \cap R_3) + P(R_1 \cap V_2 \cap R_3) \\ &= P(V_1) \cdot P(V_2 | V_1) \cdot P(R_3 | V_1 \cap V_2) + P(A_1) P(V_2 | A_1) P(R_3 | A_1 \cap V_2) \\ &\quad + P(R_1) P(V_2 | R_1) P(R_3 | R_1 \cap V_2) \end{aligned}$$

(7)

① ② ③ ④ ⑤

a)  $\frac{2! \cdot 3!}{5!}$

b)  $\frac{A_3^2 \cdot 3!}{5!}$

c)  $\frac{2 \cdot 4 \cdot 3!}{5!}$  → permutare bile

d)  $\frac{2! \cdot 3!}{5!}$

(8) cod 5 cifre

a)  $\frac{A_{10}^5}{10^5}$

b)  $\frac{5!}{10^5}$

c)  $\frac{C_5^2 \cdot 10 \cdot 9}{10^5}$

d)  $\frac{\frac{5!}{3! \cdot 1! \cdot 1!} \cdot 3 + \frac{5!}{2! \cdot 2! \cdot 1!} \cdot 3}{10^5}$

(9) 2 urne

I - 2 bile negre, 3 roșii

II - 3 bile negre, 2 roșii

1200 → impar → 2 bile I  
curat.  
2 bile II  
fără ret.

a) distribuție

Binom → cu ret.

hipergeom → fără ret.

$$P(X=k) = \frac{C_2^k \cdot \left(\frac{3}{5}\right)^k \cdot \left(\frac{2}{5}\right)^{2-k} + C_2^k \cdot C_3^{2-k}}{C_5^2}$$

$$(10) F_T(t) = \begin{cases} 0, & t < 0 \\ t/10, & t \in [0, 10] \\ 1, & t > 10 \end{cases}$$

$$16:30 \rightarrow 16:39$$

$$a) P(T \geq 9) = 1 - P(T \leq 9) = 1 - F_T(9) = 1 - \frac{9}{10} = \frac{1}{10}.$$

$$b) E(X) = \int_{-\infty}^{\infty} x f(x) dx.$$

$$f(x) = \int_{-\infty}^{+\infty} F_T(x) dx$$

$$f(x) = \begin{cases} \frac{1}{10}, & x \in [0, 10] \\ 0, & x \notin [0, 10] \end{cases}$$

$$E(X) = \int_0^{10} \frac{1}{10} \cdot x dx = \frac{1}{10} \cdot \frac{x^2}{2} \Big|_0^{10} = \frac{10}{2} = 5$$

c) 5 zile săptămână de lucru

$Z$  = nr de zile din 5 când prede bunu

$$Z \sim \text{Bino}(5, p)$$

$$\begin{aligned} P(Z \geq 2) &= 1 - P(Z=0) - P(Z=1) = 1 - C_5^0 p^0 (1-p)^5 \\ &= 1 - \left(\frac{9}{10}\right)^5 - \frac{5!}{4!} \cdot \frac{1}{10} \cdot \left(\frac{9}{10}\right)^4 \\ &= 1 - \left(\frac{9}{10}\right)^5 - \frac{1}{2} \cdot \left(\frac{9}{10}\right)^4 \end{aligned}$$

$$d) Y \sim \text{Geom}(p) = p(1-p)^{k-1}$$

$$p = P(T \geq 9) = \frac{1}{10}$$

(11)  $[1, 2] \times [2, 4] \subset \mathbb{B}^2$  se alege uniform un pct.

$(X, Y)$  coord.  $(X, Y) \sim \text{Uniform}([1, 2] \times [2, 4])$

$$f(x, y) : \mathbb{B}^2 \rightarrow \mathbb{B}$$

$$f(x, y)(x, y) = \begin{cases} 1/2, & (x, y) \in [1, 2] \times [2, 4] \\ 0, & \text{altfel} \end{cases}$$

$$(iii) f_X(x) = \begin{cases} \frac{1}{2^{-1}}, & x \in [1, 2] \\ 0, & \text{altfel} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2}, & y \in [2, 4] \\ 0, & \text{altfel} \end{cases}$$

$$X \sim \text{Unif}(1, 2) \quad Y \sim \text{Unif}(2, 4)$$

$$a) P((X, Y) \in [\frac{4}{3}, \frac{5}{3}] \times [\frac{5}{2}, \frac{4}{2}]):$$

$$P\left(\frac{4}{3} \leq X \leq \frac{5}{3}, \frac{5}{2} \leq Y \leq \frac{4}{2}\right) = \int_{\frac{4}{3}}^{\frac{5}{3}} 1 dx \cdot \int_{\frac{5}{2}}^{\frac{4}{2}} \frac{1}{2} dy$$

$$= \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$$

$$b) E(X^2 + Y^2) = E(X^2) + E(Y^2) = \int_1^2 x^2 \frac{1}{2^{-1}} dx + \int_2^4 y^2 \frac{1}{4^{-2}} dy$$

$$= \left. \frac{x^3}{3} \right|_1^2 + \frac{1}{2} \left. \frac{y^3}{3} \right|_2^4$$

$$= \frac{8-1}{3} + \frac{1}{2} \cdot \frac{64-8}{3}$$

$$= \frac{7}{3} + \frac{28}{3} = \frac{35}{3}$$

$$(14) X_1, \dots, X_m \text{ --- v.a. indep.}$$

$$P(X_i = -1) = P(X_i = 1) = 0,5$$

$$Y_i = \max\{X_i, X_{i+1}\}$$

$$1 \leq i \leq m-1$$

$$a) \text{ distr. } X_i, 1 \leq i \leq m-1$$

$$X_i, X_{i+1} \text{ v.a. indep.}$$

$$P(Y_i = -1) = P(\max\{X_i, X_{i+1}\} = -1) = P(X_i = -1, X_{i+1} = -1)$$

$$P(X_i = -1, X_{i+1} = -1) = P(X_i = -1) \cdot P(X_{i+1} = -1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\Rightarrow P(Y_i = 1) = \frac{3}{4}$$

$$Y_i \sim \begin{pmatrix} -1 & 1 \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

$$b) E(Y_i) = \frac{1}{4} \cdot (-1) + \frac{3}{4} = \frac{2}{4} = \frac{1}{2} \quad V(Y) = 1 - \left(\frac{1}{2}\right)^2$$

$$V(X) = E(X^2) - E^2(X) = \frac{3}{4}$$

$$c) Z_n = \frac{1}{n} (X_1^5 + \dots + X_n^5)$$

$$\angle TNM \Rightarrow E(X_1^5) = 0$$

$$(Z_n)_n \xrightarrow{a.s.} E(X_1^5) = 0$$

$$\frac{1}{2} \cdot (-1)^5 + \frac{1}{2} = 0$$

$$(15) \quad 10 \text{ min} \quad X \sim \text{Exp}(\lambda)$$

$$a) P(T \leq 15) = ?$$

$$b) E(X) = ? \quad P(T > 15) = 0,1$$

$$E(X) = \frac{1}{\lambda}$$

$$E(X) = 10 \Rightarrow \lambda = \frac{1}{10}$$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$f_X(x) = \begin{cases} \frac{1}{10} \cdot e^{-\frac{1}{10}x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$P(T \leq 15) = \int_0^{15} \lambda e^{-\lambda t} dt = \frac{1}{10} \int_0^{15} e^{-\frac{t}{10}} dt$$

$$\frac{1}{10} \cdot \left( -\frac{10}{e^{t/10}} \right) \Big|_0^{15}$$

$$= -\frac{1}{e^{15/10}} + 1 = 1 - e^{-1.5}$$

$$b) P(T > 15) = 1 - P(T \leq 15) = e^{-15 \cdot 2} \rightarrow e^{-2 \cdot 15} = 0,1$$

$$\hookrightarrow 0,1$$

$$-2 \cdot 15 = \ln \frac{1}{10}$$

$$E(X) = \frac{1}{\lambda} = \frac{15}{\ln 10}$$

Cens final

$$(1) U = [1, 2, 3, 1, 2, 4, 1, 2, 2, 4]$$

vector de date albe  $\rightarrow$  cu ret.  $U: X = [U_{i1}, \dots, U_{i5}]$

$\searrow$  fără ret.  $U: Y = [U_{j1}, U_{j2}, U_{j3}]$

2. v. a. indice de cost ori apare 1 ori x

$$a) \quad \begin{array}{l} 1: 3 \\ 2: 4 \\ 3: 1 \\ 4: 2 \end{array}$$

$$(3) f: \mathbb{R} \rightarrow [0, \infty)$$

$$f(x) = \begin{cases} \frac{2x}{a^2}, & x \in [0, a] \\ 0, & \text{and elsewhere} \end{cases}$$

$$a) F_X(x) = ? \quad P(X < \frac{a}{2})$$

$$P(|X - \frac{a}{2}| < \frac{a}{4})$$

$$a) F_X(x) = \int_{-\infty}^x f(t) dt = \begin{cases} \int_0^x \frac{2t}{a^2} dt & x \in [0, a] \\ 0, & \text{and elsewhere} \end{cases} = \begin{cases} 1/a, & x \in [0, a] \\ 0, & \text{and elsewhere} \end{cases}$$

$$\int_0^a \frac{2x}{a^2} dx = \frac{2}{a^2} \cdot \int_0^a x dx = \frac{2}{a^2} \cdot \left. \frac{x^2}{2} \right|_0^a = \frac{x^2}{a^2} \Big|_0^a = \frac{1}{a}$$

$$P(X < \frac{a}{2}) = \int_0^{\frac{a}{2}} \frac{1}{a} dx = \left. \frac{x}{a} \right|_0^{\frac{a}{2}} = \frac{a}{2} \cdot \frac{1}{a} = \frac{1}{2}$$

$$P(|X - \frac{a}{2}| < \frac{a}{4}) = P(X < \frac{3a}{4}, X > \frac{a}{4}) = P(X < \frac{3a}{4}) \cdot P(X > \frac{a}{4})$$

$$|X - \frac{a}{2}| < \frac{a}{4} \rightarrow X - \frac{a}{2} < \frac{a}{4} \Rightarrow X < \frac{3a}{4} \quad = \frac{3}{16}$$

$$\frac{a}{2} - X < \frac{a}{4} \Rightarrow \frac{a}{4} < X$$

$$P(X < \frac{3a}{4}) = \int_0^{\frac{3a}{4}} \frac{1}{a} dx = \left. \frac{x}{a} \right|_0^{\frac{3a}{4}} = \frac{3a}{4} \cdot \frac{1}{a} = \frac{3}{4}$$

$$P(X > \frac{a}{4}) = 1 - P(X \leq \frac{a}{4}) = 1 - \int_0^{\frac{a}{4}} \frac{1}{a} dx = 1 - \left. \frac{x}{a} \right|_0^{\frac{a}{4}} = 1 - \frac{1}{4} = \frac{3}{4}$$

$$(4) f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(t) = \begin{cases} \frac{1}{d}, & t \in [a, a+d] \\ 0, & t \notin [a, a+d] \end{cases}$$

$$V(T) = \frac{3}{4}$$

$$a) E(T) = ?, E(T^2) = ?$$

$$b) d = ?$$

$$V(T) = \frac{3}{4}$$

$$F_T(t) = \begin{cases} \int_a^{a+d} \frac{1}{d} dt = \frac{a+d - a}{d} = 1 \\ 0, \text{ anders} \end{cases}$$

$$V(T) = E(T^2) + E^2(T)$$

$$E(T) = \int_a^{a+d} t \cdot \frac{1}{d} dt = \frac{t^2}{2d} \Big|_a^{a+d} = \frac{(a+d)^2 - a^2}{2d} \\ = \frac{2ad + d^2}{2d} = \frac{2a+d}{2}$$

$$E(T^2) = \int_a^{a+d} t^2 \cdot \frac{1}{d} dt = \frac{t^3}{3d} \Big|_a^{a+d}$$

$$(a+d)^3 = (a+d)^2 (a+d) = (a^2 + 2ad + d^2)(a+d)$$

$$= a^3 + a^2d + 2a^2d + 2ad^2 + ad^2 + d^3$$

$$= a^3 + 3a^2d + 3ad^2 + d^3$$

$$E(T^2) = \frac{a^3}{3d} + \frac{3a^2d}{3d} + \frac{3ad^2}{3d} + \frac{d^3}{3d} - \frac{a^3}{3d}$$

$$= \frac{a^3}{3d} + \frac{d^2}{3} - \frac{a^3}{3d} + a^2 + ad = a^2 + ad + \frac{d^2}{3}$$

$$b) \frac{3}{4} = a^2 + a\alpha + \frac{\alpha^2}{3} - \left(a + \frac{\alpha}{2}\right)^2$$

$$\frac{3}{4} = \cancel{a^2} + \cancel{a\alpha} + \frac{\alpha^2}{3} - \cancel{a^2} - \cancel{a\alpha} - \frac{\alpha^2}{4} \Leftrightarrow \frac{3}{4} = \frac{\alpha^2}{3} - \frac{\alpha^2}{4} \Leftrightarrow \alpha^2 = 3 \Leftrightarrow \alpha = 3$$

(5)  $X_i \sim \text{Bernoulli}(p_i)$ ,  $i \in \{1, 2, 3\}$  v.a. independ.

$X_i = 1$   $R_i$  funkt.

$X_i = 0$  :  $R_i$  mefunkt.

$P = ?$

$$X \sim \begin{pmatrix} 1 & 2 & 3 \\ 0,8 & 0,7 & 0,9 \end{pmatrix}$$

$$P((R_1 \cap R_3) \cup B_2)$$

$$= P(R_1 \cap R_3) + P(B_2) - P((R_1 \cap R_3) \cap B_2)$$

$$= P(R_1) \cdot P(R_3) + P(B_2) - P(R_1) P(R_2) P(R_3)$$

$$= 0,8 \cdot 0,9 + 0,7 - 0,8 \cdot 0,7 \cdot 0,9$$

$$= 1,42 - 0,50 = 0,92$$

$$(6) \quad P(T > 2)$$

$$P(T = 2)$$

$$P(T < 2)$$

$$P(T < 2) = p_1 \cdot P(T_1 < 2) + p_2 \cdot P(T_2 < 2)$$

$$+ p_3 \cdot P(T_3 < 2)$$

$$f_{T_1}(T_1) = \begin{cases} \frac{1}{4}, & x \in [1, 5] \\ 0, & \text{andere} \end{cases}$$

$$f_{T_2}(T_2) = \begin{cases} \frac{1}{2}, & T_2 \in [1, 3] \\ 0, & \text{andere} \end{cases}$$

$$f_{T_3}(T_3) = \begin{cases} \frac{1}{3}, & x \in [1, 4] \\ 0, & \text{andere} \end{cases}$$

$$T_1: \int_1^2 \frac{1}{4} dt = \left. \frac{t}{4} \right|_1^2 = \frac{1}{4}$$

$$T_2: \int_1^2 \frac{1}{2} dt = \left. \frac{t}{2} \right|_1^2 = \frac{1}{2}$$

$$T_3: \int_1^2 \frac{1}{3} dt = \left. \frac{t}{3} \right|_1^2 = \frac{1}{3}$$

$$P(T < 2) = 0,4 \cdot \frac{1}{4} + 0,4 \cdot \frac{1}{2} + 0,2 \cdot \frac{1}{3} = 0,1 + 0,2 + 0,06 = 0,36$$

(7) 4 bilete cãștig  
3 bilete f. returnare  
8 bilete mecatig.

a)  $p(4 \text{ m. - or extras bilete } e^4)$

$$X \sim \begin{pmatrix} 1 & 2 & 3 \\ \frac{8}{12} & \frac{4}{11} & \frac{6}{10} \end{pmatrix}$$

$$\frac{\frac{2}{8 \cdot 7 \cdot 6 \cdot 5} \cdot 3}{\frac{12 \cdot 11 \cdot 10}{5}} = \frac{14}{55}$$

b)  $X$  v.a. m. bilete câștig. extras

$$E(X) = \sum_{i=1}^m x_i \cdot p_i$$

Subiect 25.01.2023

(i) multimea  $\{0, 1, 2, 3, 4, 5, 6\}$  coded  $X$  - 4 cifre distincte

a) cazuri posibile:  $7 \cdot 6 \cdot 5 \cdot 4$

b)  $X = 1234$  sau  $X = 0246 \rightarrow P = \frac{2}{7 \cdot 6 \cdot 5 \cdot 4}$

c)  $X = 0123, 1234, 2345, 3456$   
 $P = \frac{4}{7 \cdot 6 \cdot 5 \cdot 4}$

d)  $X \rightarrow$  doar cifre pare  $P = \frac{4!}{7 \cdot 6 \cdot 5 \cdot 4}$

e)  $X = \dots 4$   $P = \frac{A_6^3 \cdot 3!}{4 \cdot 5 \cdot 6 \cdot 7}$

f)  $1 - P(\bar{F}) = 1 - \frac{4!}{4 \cdot 5 \cdot 6 \cdot 7}$

$$(3) \quad f(x) = \begin{cases} \frac{1}{2}, & x \in [a, a+2] \\ 0, & \text{otherwise} \end{cases}$$

$$a) \quad P(X \leq a) \quad P(X \leq a + \frac{1}{2})$$

$$P(X \leq a) = \int_a^a \frac{1}{2} dx = \int_{-\infty}^a \frac{1}{2} dx = 0.$$

$$\begin{aligned} P(X \leq a + \frac{1}{2}) &= \int_a^{a+\frac{1}{2}} \frac{1}{2} dx = \int_a^a \frac{1}{2} dx + \int_a^{a+\frac{1}{2}} \frac{1}{2} dx \\ &= \left. \frac{x}{2} \right|_a^{a+\frac{1}{2}} = \frac{a+\frac{1}{2}}{2} - \frac{a}{2} = \frac{\frac{1}{2}}{2} = \frac{1}{4} \end{aligned}$$

$$F(x) = \int_{-\infty}^{a+2} f(x) dx = \begin{cases} 1, & x \geq a \\ 0, & x < a \end{cases}$$

$$\lim_{x \rightarrow a} F(x) = \lim_{x \rightarrow a} 1 = 1.$$

$$F(a + \frac{1}{2}) = \frac{1}{4}.$$